

Section : Physics

Q.1 A long wire is bent into a circular coil of one turn and then into a circular coil of smaller radius having n turns. If the same current passes in both the cases, then the ratio of the magnetic fields produced at the centers in two cases will be:

1. $1 : n^2$
2. $1 : n$
3. $\sqrt{n} : 1$
4. $1 : n^3$

Options 1. 1
2. 2
3. 3
4. 4

Q.2 A charging capacitor has
1. Only conduction current present in connecting wire
2. Only displacement current present in the space between plates
3. Both displacement and conduction currents present
4. No current present in different regions of space

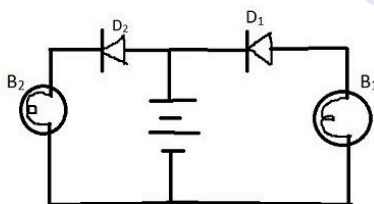
Options 1. 1
2. 2
3. 3
4. 4

Q.3 Two point charges $q_1 = 36 \mu\text{C}$ and $q_2 = -9 \mu\text{C}$ are placed at a distance of 30 cm. The distance from q_1 where the net electric field is zero, will be:

1. 10 cm
2. 20 cm
3. 60 cm
4. 30 cm

Options 1. 1
2. 2
3. 3
4. 4

Q.4 In the circuit diagram, the battery, diodes D_1 and D_2 , bulbs B_1 and B_2 are connected as shown. Which of the following statements is true regarding bulbs B_1 and B_2 ?



1. Only B_1 will glow
2. B_1 and B_2 both will glow
3. Only B_2 will glow
4. Both B_1 and B_2 will not glow

Options 1. 1
2. 2
3. 3
4. 4

Q.5 Two Polaroids are placed in such a way that their axes make an angle of 60° with each other. If an unpolarized light is incident on a system of these Polaroids, the fraction of transmitted light to the incident light will be

1. zero
2. $1/8$
3. $3/8$
4. $5/8$

Options 1. 1
2. 2
3. 3

- Q.6** If we double the number of turns of coil in a moving coil galvanometer, its voltage sensitivity will:
1. reduce to half
 2. remain unchanged
 3. increase to twice
 4. increase to 4 times

Options

1. 1
2. 2
3. 3
4. 4

- Q.7** An iron ring of relative permeability μ_r is introduced inside a toroidal solenoid having η turns per meter. If the current in the solenoid is I , then the magnetic field in the ring is:

1. $\frac{\mu_0 \mu_r I}{\sqrt{\eta}}$
2. $\frac{\mu_0 \mu_r I}{\eta^2}$
3. $\mu_0 \mu_r \eta^2 I$
4. $\mu_0 \mu_r \eta I$

Options

1. 1
2. 2
3. 3
4. 4

- Q.8** Match List-I with List-II

List-I

List-II

(A) E is independent of r (I) For a point charge

(B) $E \propto \frac{1}{r}$

(II) For a short dipole

(C) $E \propto \frac{1}{r^2}$

(III) For a linear charge distribution

(D) $E \propto \frac{1}{r^3}$

(IV) For a surface charge distribution on an infinite plane sheet

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)
3. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)
4. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

Options

1. 1
2. 2
3. 3
4. 4

- Q.9** Three resistors of resistance 15Ω each are to be connected in combination. The ratio of maximum equivalent resistance to the minimum equivalent resistance is:

1. 3:1
2. 6:1
3. 9:1
4. 12:1

Options

1. 1
2. 2
3. 3
4. 4

- Q.10** Match List-I with List-II

List-I

List-II

Potential difference across the terminal of a battery

Situations

(A) $V = E$

(I) Cell is being charged

(B) $V = 0$

(II) The internal resistance of a cell is zero

(C) $V = E - ir$

(III) cell is short-circuited

(D) $V = E + ir$

(IV) current is being drawn from a cell

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)
2. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
3. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)
4. (A) - (III), (B) - (II), (C) - (IV), (D) - (I)

Options

1. 1
2. 2
3. 3
4. 4

Q.11 Interference fringes may be observed due to the superposition of two light waves represented by:

1. $y_1 = a \sin(\omega t)$, $y_2 = a \sin(2\omega t)$
2. $y_1 = a \sin(\omega t)$, $y_2 = 2a \sin(2\omega t + \delta)$
3. $y_1 = 2a \sin(2\omega t)$, $y_2 = 2a \sin(2(\omega t + \delta))$
4. $y_1 = 2a \sin(\omega t)$, $y_2 = 2a \sin(2\omega t + \delta)$

Options

1. 1
2. 2
3. 3
4. 4

Q.12 A long straight wire of radius 'a' carries a steady current 'I'. The current is uniformly distributed across its area of cross-section. The ratio of magnitude of magnetic fields B₁ at a distance a/2 and B₂ at a distance 2a from the axis of the wire.

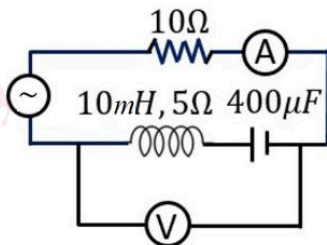
1. 1:2
2. 1:1
3. 2:1
4. 4:1

Options

1. 1
2. 2
3. 3
4. 4

Q.13

In the circuit shown in the figure, the ac source gives a voltage of $V = 10 \cos(500t)$. The ammeter reading will be:



1. $\frac{2\sqrt{2}}{3}$ A
2. $\frac{3\sqrt{2}}{3}$ A
3. $\frac{\sqrt{2}}{3}$ A
4. $\sqrt{2}$ A

Options

1. 1
2. 2
3. 3
4. 4

Q.14 If h is the Planck's constant, then the momentum of the photon having frequency ν is represented by

1. hcv
2. $h\nu/c$
3. hc/ν
4. $h/c\nu$

Options

1. 1
2. 2
3. 3
4. 4

Q.15 A symmetrical double convex lens is cut into two equal parts by a plane containing the principal axis:

The power of the divided lens is:

1. Half of the original lens
2. equal to that of the original lens

3. double of the original lens
4. four times of the original lens

Options 1. 1

2. 2
3. 3
4. 4

Q.16 When white light consisting of wavelengths ranging from 400 nm to 700 nm passes through water ($\mu = 4/3$), then the wavelength range in water will be:

1. 300 nm - 525 nm
2. 500 nm - 700 nm
3. 700 nm - 850 nm
4. 450 A -750 A

Options 1. 1

2. 2
3. 3
4. 4

Q.17 Light of wavelength 300 nm is used in an experiment on photoelectric effect with lithium having a work function of 2.7 eV. The maximum kinetic energy of ejected photo-electrons will be

1. 1.60 eV
2. 1.90 eV
3. 1.75 eV
4. 1.43 eV

Options 1. 1

2. 2
3. 3
4. 4

Q.18 An infinitely long line charge distribution produces an electric field of 9×10^4 N/C at a distance of 2 cm. The linear charge density of the distribution is:

1. $0.01 \mu\text{C/m}$
2. $0.1 \mu\text{C/m}$
3. $1 \mu\text{C/m}$
4. $2 \mu\text{C/m}$

Options 1. 1

2. 2
3. 3
4. 4

Q.19 A cell of emf E is connected across an external resistance R . When a current I is drawn, the potential difference across the electrodes drops to V . The internal resistance r of the cell is:

1. $\left[\frac{E - V}{E}\right] R$
2. $\left[\frac{E - V}{R}\right]$
3. $\left[\frac{E - V}{I}\right] R$
4. $\left[\frac{E - V}{V}\right] R$

Options 1. 1

2. 2
3. 3
4. 4

Q.20 An equiconvex lens has a focal length $\frac{2}{3}$ times the radius of curvature of either surface. The refractive index of the material of the lens is:

1. 1.29
2. 1.52
3. 1.63
4. 1.75

Options 1. 1

2. 2

3. 3

4. 4

Q.21 The smallest wavelength in the spectral lines of the Paschen series of hydrogen atom is:
(R is the Rydberg's constant in AA^{-1})

1. $\frac{9}{R} \text{AA}$
2. $\frac{R}{9} \text{AA}$
3. $\frac{9R}{21} \text{AA}$
4. $\frac{21}{R} \text{AA}$

Options

1. 1
2. 2
3. 3
4. 4

Q.22 The drift velocity of electrons in a conductor is ' v ' when a current ' i ' flows through it. If the area of the cross-section of the conductor is reduced to half of its initial value, then the drift velocity will be:

1. $v/2$
2. v
3. $2v$
4. $4v$

Options

1. 1
2. 2
3. 3
4. 4

Q.23 The external diameter of a 5 m long hollow copper tube is 10 cm and the wall thickness is 5 mm. If the specific resistance of copper is $1.7 \times 10^{-8} \Omega\text{m}$, the resistance of the tube will be:

1. $7.45 \times 10^{-4} \Omega$
2. $14.9 \times 10^{-4} \Omega$
3. $5.7 \times 10^{-5} \Omega$
4. $5.7 \times 10^5 \Omega$

Options

1. 1
2. 2
3. 3
4. 4

Q.24 A uniform electric field exists in the positive x-direction. Let V_A = electric potential at (0, 0, 0) cm V_B = electric potential at (5, 0, 0) cm and V_C = electric potential at (0, 5, 0) cm. The correct relationship between them is:

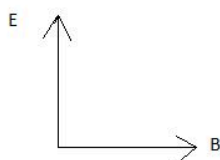
1. $V_A < V_C$ and $V_A = V_B$
2. $V_A = V_C$ and $V_A > V_B$
3. $V_A > V_C$ and $V_A = V_B$
4. $V_A = V_C$ and $V_A < V_B$

Options

1. 1
2. 2
3. 3
4. 4

Q.25

From the following diagram, determine the direction of propagation of an electromagnetic wave.



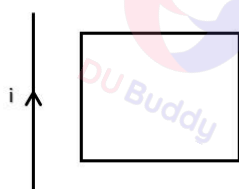
1. In the direction of electric field
2. In the direction of magnetic field
3. Inward to the plane of paper
4. Outward from the plane of paper

Options

1. 1
2. 2
3. 3
4. 4

Q.26

A square loop is placed near a long straight current carrying wire as shown in figure. If the loop is moved away from the wire



1. Induced current in loop will be anti-clockwise
2. Wire will exert net attractive force on the loop
3. Wire will exert net repulsive force on the loop
4. No current is induced in the loop

Options

1. 1
2. 2
3. 3
4. 4

Q.27

A short bar magnet of magnetic moment $5.25 \times 10^{-2} \text{ J/T}$ is placed with its axis perpendicular to Earth's magnetic field direction. The distance from the center of the magnet on its perpendicular bisector where the resultant field makes 45° with Earth's field will be:
(Earth's magnetic field at that place is 0.42 G .)

1. $6.3 \times 10^{-2} \text{ m}$
2. $5 \times 10^{-2} \text{ m}$
3. $6.3 \times 10^{-3} \text{ m}$
4. $5 \times 10^{-3} \text{ m}$

Options

1. 1
2. 2
3. 3
4. 4

Q.28 In an intrinsic semiconductor at room temperature;

1. the number of free electrons are much larger than the number of holes
2. the number of holes are much larger than the number of free electrons
3. the number of free electrons and the number of holes are equal and non-zero
4. the number of free electrons and the number of holes are zero

Options

1. 1
2. 2
3. 3
4. 4

Q.29

In a series LCR circuit, $C = 5 \mu\text{F}$ and $\omega = 1000 \text{ rad/s}$.

The value of inductance L for which the current is maximum in this circuit will be:

1. 100 mH

2. 10 mH

3. 50 mH

4. 200 mH

Options 1. 1

2. 2

3. 3

4. 4

Q.30 If n_i is the concentration of intrinsic charge carriers in a pure semiconductor and n_e and n_h be the concentrations of electrons and holes, respectively after the semiconductor is doped, then at thermal equilibrium.

1. $n_e n_h = n_i$

2. $n_e n_h = n_i^2$

3. $\frac{n_e}{n_h} = n_i^2$

4. $\frac{n_h}{n_e} = n_i^2$

Options 1. 1

2. 2

3. 3

4. 4

Q.31 The element having the highest binding energy per nucleon among the following is:

1. ${}^6\text{Li}$

2. ${}^{14}\text{N}$

3. ${}^{12}\text{C}$

4. ${}^{16}\text{O}$

Options 1. 1

2. 2

3. 3

4. 4

Q.32 A ray of light from air passes through an equilateral triangular glass prism and undergoes minimum deviation when the angle of incidence is $\frac{3}{4}$ of the angle of prism. The speed of light in the prism is:

1. $1.15 \times 10^8 \text{ m/s}$

2. $2.12 \times 10^8 \text{ m/s}$

3. $2.78 \times 10^8 \text{ m/s}$

4. $3 \times 10^8 \text{ m/s}$

Options 1. 1

2. 2

3. 3

4. 4

Q.33 If the ratio of mass number of two nuclei is 8:27, then the ratio of the volumes of their nuclei is:

1. 8:27

2. 4:9

3. 2:3

4. 1:1

Options 1. 1

2. 2

3. 3

4. 4

Q.34 In a plane electromagnetic wave, the electric field varies as $E = 90 \sin(500x + 20 \times 10^{10}t) \text{ V/m}$.

The expression for the magnetic field variation is:

1. $27 \times 10^{-7} \sin(500x + 20 \times 10^{10}t) \text{ T}$

2. $3.3 \times 10^6 \sin(500x + 20 \times 10^{10}t) \text{ T}$

3. $27 \times 10^8 \sin(500x + 20 \times 10^{10}t) \text{ T}$

4. $3 \times 10^{-7} \sin(500x + 20 \times 10^{10}t) \text{ T}$

Options 1. 1

2. 2

3. 3

4. 4

Q.35 A current of 15 A flows in a wire of cross-sectional area 5 mm^2 with a drift velocity of $3 \times 10^{-3} \text{ m/s}$. The electron density in the wire will be:

1. $489 \times 10^{25} \text{ m}^{-3}$

2. $546 \times 10^{25} \text{ m}^{-3}$

3. $725 \times 10^{23} \text{ m}^{-3}$

4. $625 \times 10^{25} \text{ m}^{-3}$

Options 1. 1

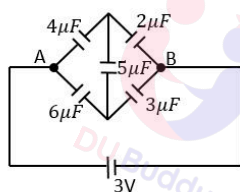
2. 2

3. 3

4. 4

Q.36

A network of five capacitors is shown in the figure. The total charge stored in the network connected between A and B is:



1. $8 \mu\text{C}$
2. $10 \mu\text{C}$
3. 9 mC
4. 10 mC

Options 1. 1

2. 2

3. 3

4. 4

Q.37 A negatively charged object A is repelled by another charged object B. However, an object C is attracted to object B.

Which of the following is the most probable for the charge of object C?

1. neutral or negatively charged
2. negatively charged only
3. neutral or positively charged
4. positively charged only

Options 1. 1

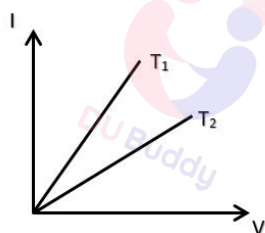
2. 2

3. 3

4. 4

Q.38

The current-voltage graphs for a given metallic wire at two different temperatures, T_1 and T_2 are shown in the figure. The correct relation of temperatures and resistances will be :



1. $R_{T_2} = R_{T_1}, T_1 > T_2$
2. $R_{T_2} < R_{T_1}, T_1 > T_2$
3. $R_{T_2} > R_{T_1}, T_2 > T_1$
4. $R_{T_2} < R_{T_1}, T_2 > T_1$

Options 1. 1

2. 2

3. 3

Q.39 A monochromatic source emitting light of wavelength 600 nm has a power output of 66 W.

The number of photons emitted by the source in 2 minutes is: (Given: $h = 6.6 \times 10^{-34}$ Js)

1. 1.2×10^{22}
2. 2.4×10^{22}
3. 12×10^{22}
4. 24×10^{22}

Options

1. 1
2. 2
3. 3
4. 4

Q.40 A body of mass 20 g and charge $+50 \mu\text{C}$ is suspended from the ceiling by a string.

A horizontal electric field of 10 kV/m is applied.

The angle between the string and the vertical direction is: (Take $g = 10 \text{ N/kg}$).

1. $\tan^{-1}(2.50)$
2. $\tan^{-1}(3.00)$
3. $\tan^{-1}(3.50)$
4. $\sin^{-1}(2.50)$

Options

1. 1
2. 2
3. 3
4. 4

Q.41 The magnifying power of a compound microscope is high, if

1. objective has long focal length and eyepiece has short focal length
2. objective has short focal length and eyepiece has long focal length
3. Both objective and eyepiece have short focal length
4. Both objective and eyepiece have long focal length

Options

1. 1
2. 2
3. 3
4. 4

Q.42 A long solenoid with 30 turns/cm has a small loop of area 3 cm^2 placed inside it normal to its axis.

If the current in the solenoid changes steadily from 3.0 A to 6.0 A in 0.1 s, the induced emf in the loop is:

1. $3\pi \times 10^{-7} \text{ V}$
2. $9\pi \times 10^{-7} \text{ V}$
3. $18\pi \times 10^{-7} \text{ V}$
4. $108\pi \times 10^{-7} \text{ V}$

Options

1. 1
2. 2
3. 3
4. 4

Q.43 A coil having area A is placed in a magnetic field. The field is perpendicular to the plane of the coil.

When the magnetic field changes from B_0 to $4B_0$ in a time interval t ,

the magnitude of emf induced in the coil will be:

1. $\frac{3A_o B_0}{t}$
2. $\frac{4A_o B_0}{t}$
3. $\frac{3B_0}{A_o t}$
4. $\frac{4B_0}{A_o t}$

Options

1. 1
2. 2
3. 3

Q.44 The electric field in a region is $\vec{E} = (6\hat{i} + 3\hat{j} + 5\hat{k}) \text{ N/C}$.

The electric flux due to this electric field through an area of 100 cm^2 lying in the X-Y plane is:

1. $5 \times 10^2 \text{ N m}^2/\text{C}$
2. $5 \times 10^1 \text{ N m}^2/\text{C}$
3. $5 \times 10^{-1} \text{ N m}^2/\text{C}$
4. $5 \times 10^{-2} \text{ N m}^2/\text{C}$

Options 1. 1

2. 2
3. 3
4. 4

Q.45 The current sensitivity of a galvanometer increases by 10%. If its resistance also increases by 20%, the voltage sensitivity will

1. decrease by 8.3 %
2. increase by 8.3 %
3. decrease by 21.5%
4. remain unaffected

Options 1. 1

2. 2
3. 3
4. 4

Q.46 Young's double slit experiment is carried out using green, yellow, red, and blue lights one by one. Arrange the expected values of fringe widths (β) in increasing order:

- (A) β_{Green}
- (B) β_{Yellow}
- (C) β_{Blue}
- (D) β_{Red}

Choose the correct answer:

1. (B), (C), (D), (A)
2. (A), (C), (D), (B)
3. (C), (A), (B), (D)
4. (C), (A), (D), (B)

Options 1. 1

2. 2
3. 3
4. 4

Q.47 The mutual inductance of two co-axial coils is 2 H. If the current in one coil changes uniformly from zero to 1.0 A in 100 ms, the emf induced in the other coil is

1. 10 V
2. 20 V
3. -10 V
4. -20 V

Options 1. 1

2. 2
3. 3
4. 4

Q.48 The atomic hydrogen emits a line spectrum consisting of different series. The ratio of frequencies of the first line in Lyman and Balmer series is:

1. 3:5
2. 9:5
3. 27:5
4. 5:27

Options 1. 1

2. 2
3. 3
4. 4

Q.49 Match List-I with List-II

List-I

List-II

Motion of a charge particle

Path

- (A) Initial velocity of charge particle parallel to B (I) Circular Path
(B) Initial velocity of charge particle perpendicular to B (II) Helical Path
(C) Initial velocity of charge particle inclined to B (III) Straight Path
(D) Initial velocity of charge particle perpendicular to E (IV) Parabolic Path

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
3. (A) - (III), (B) - (I), (C) - (II), (D) - (IV)
4. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

Options

1. 1
2. 2
3. 3
4. 4

Q.50 Match List-I with List-II

List-I

List-II

AC circuit

Features

- (A) Only resistor in a circuit (I) Current leads or lag the voltage
(B) Only capacitor in a circuit (II) Voltage and Current are in phase
(C) Only inductor in a circuit (III) The current lags the voltage by $\frac{\pi}{2}$
(D) Series LCR circuit (IV) The current leads the voltage by $\frac{\pi}{2}$
 $X_C \neq X_L$ and $R \neq 0$

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
2. (A) - (II), (B) - (IV), (C) - (III), (D) - (I)
3. (A) - (III), (B) - (II), (C) - (IV), (D) - (I)
4. (A) - (IV), (B) - (II), (C) - (III), (D) - (I)

Options

1. 1
2. 2
3. 3
4. 4